

## 1. Input Data & Normalization

Raw KT Logs (ASSISTments 2012, Junyi, XES3G5M) → Schema Standardization

## 2. Splitting & Stratification

Learner-based / Temporal Splits → Train-only KC-frequency Buckets

## 3. Baseline Evaluation

Model Training (IRT, DKT, SimpleKT) → Raw Prediction CSV Export

## 4. Diagnostic Metrics & Stratification Suite

- Aggregate Metrics: AUC, Accuracy, NLL, RMSE
- Calibration Strata: Expected Calibration Error (ECE) & Brier Decomposition
  - Visual Analytics: Multi-seed Reliability Diagrams per KC Bucket
- Cold-start Subsets: Strict ( $f_{\text{train}} = 0$ ),  $k$ -shot subsets ( $f_{\text{train}} \leq 5$  or  $\leq 10$ ) & Warm cohorts

## 5. Hygiene Audit & Reproducibility

7-Channel Leakage Checklist Audit (L1-L7)

Automatically Generated Reports → Standardized LaTeX/PDF Artifacts